

CO4-AT1-Q2:

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Course Name: DAA for Complexity Analysis

Huffman Coding Minimum Cost Problem Statement

Huffman Coding is a lossless data compression technique that assigns shorter codes to frequently occurring characters and longer codes to less frequent characters.

The objective is to calculate the minimum weighted path length (encoding cost).

Problem Understanding

The algorithm combines the two smallest frequencies repeatedly to create an optimal binary tree.

Example:

Frequencies:

5, 9, 12, 13, 16

Minimum cost obtained:

224

Algorithm Used

Huffman Coding

Steps:

1. Arrange frequencies in ascending order.
2. Select two smallest frequencies.
3. Combine them.
4. Add the combined value to total cost.
5. Repeat until one node remains.

Time Complexity

Sorting approach:

$O(n^2)$

(Using priority queue)

$O(n \log n)$

Space Complexity

$O(n)$

Result

Huffman Coding successfully generates minimum encoding cost by assigning efficient codes and reducing storage requirements.